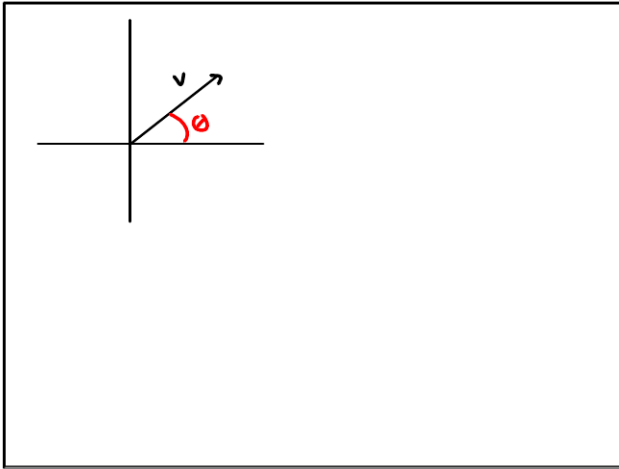
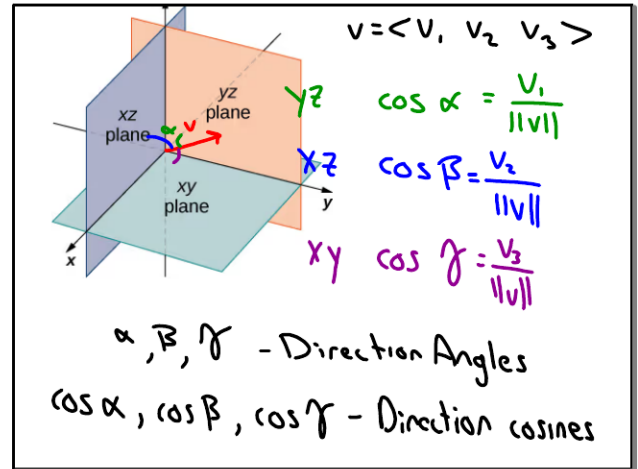


Ameet Rao's Calculus III Notes

Direction Angles and Cosines, Determinants, and the Cross Product



A vector v makes an angle θ with the x -axis. Direction angles carry this idea into 3D.



A 3D vector makes three angles with the axes: α (x), β (y), γ (z). Their cosines are the direction cosines: $\cos \alpha = v_1/\|v\|$, $\cos \beta = v_2/\|v\|$, $\cos \gamma = v_3/\|v\|$.

$$2i + 3j + 4k$$

$$\cos \alpha = \frac{2}{\sqrt{29}} \quad \alpha = 1.190 \quad 69.2^\circ$$

$$\cos \beta = \frac{3}{\sqrt{29}} \quad \beta = 0.98 \quad 56.15^\circ$$

$$\cos \gamma = \frac{4}{\sqrt{29}} \quad \gamma = 0.734 \text{ rad} \quad 42.05^\circ$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\left(\frac{v_1}{\|v\|}\right)^2 + \left(\frac{v_2}{\|v\|}\right)^2 + \left(\frac{v_3}{\|v\|}\right)^2 = 1$$

$$\frac{v_1^2 + v_2^2 + v_3^2}{v_1^2 + v_2^2 + v_3^2} = 1$$

For $2i + 3j + 4k$, $\|v\| = \sqrt{29}$ gives $\alpha \approx 1.19$, $\beta \approx 0.98$, $\gamma \approx 0.73$ rad. Squared direction cosines always sum to 1.

determinant

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \det(A) = ad - bc$$

A 2×2 determinant multiplies down the main diagonal and subtracts the other diagonal: $\det(A) = ad - bc$. It is the building block of the cross product.

Cross Product

$$u = \langle u_1, u_2, u_3 \rangle$$

$$v = \langle v_1, v_2, v_3 \rangle \quad u \times v$$

$$\begin{bmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{bmatrix} \begin{aligned} (i) \det \begin{bmatrix} u_2 & u_3 \\ v_2 & v_3 \end{bmatrix} &= (u_2 v_3 - u_3 v_2) i \\ (-j) \det \begin{bmatrix} u_1 & u_3 \\ v_1 & v_3 \end{bmatrix} &= -(u_1 v_3 - u_3 v_1) j \\ (k) \det \begin{bmatrix} u_1 & u_2 \\ v_1 & v_2 \end{bmatrix} &= (u_1 v_2 - u_2 v_1) k \end{aligned}$$

$$u \times v = (u_2 v_3 - u_3 v_2)i - (u_1 v_3 - u_3 v_1)j + (u_1 v_2 - u_2 v_1)k$$

The cross product $u \times v$ expands a 3×3 determinant along the top row with signs $+$, $-$, $+$. The result is a vector perpendicular to both u and v .

$$u = i - 2j + k$$

$$v = 3i + j - 2k$$

$$u \times v = 3i + 5j + 7k$$

$$v \times u = -3i - 5j - 7k$$

$u = i - 2j + k$, $v = 3i + j - 2k$ gives $u \times v = 3i + 5j + 7k$. Order matters: $v \times u = -3i - 5j - 7k$, the exact opposite.

Ameet Rao's Calculus III Notes

Direction Angles and Cosines, Determinants, and the Cross Product

$$\mathbf{v} \times \mathbf{v} = \mathbf{0}$$

$$\mathbf{0} \times \mathbf{v} = \mathbf{0}$$

$$\mathbf{v} \times (c\mathbf{v}) = \mathbf{0}$$

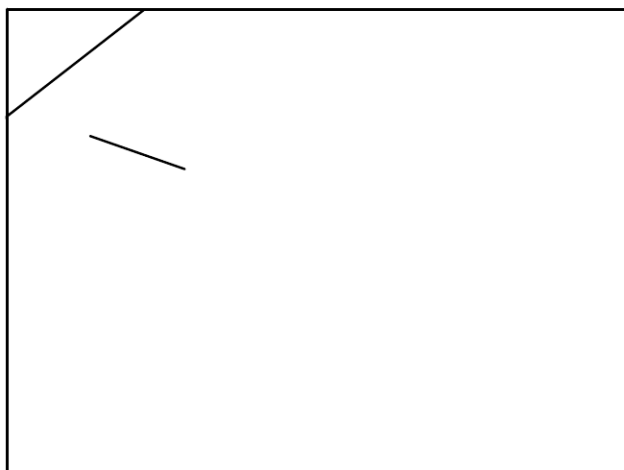
$$\mathbf{v} \times (c\mathbf{v}) = \mathbf{0} \quad c \in \mathbb{R}$$

Crossing a vector with itself or with any scalar multiple of itself gives the zero vector: $\mathbf{v} \times \mathbf{v} = \mathbf{0}$, $\mathbf{v} \times c\mathbf{v} = \mathbf{0}$.

Two **vectors** must not be
scalar multiples

If they are $\mathbf{v} \times c\mathbf{v} = \mathbf{0}$

For a useful cross product the two vectors can't be scalar multiples. If they are (parallel), $\mathbf{v} \times c\mathbf{v} = \mathbf{0}$, so no perpendicular direction exists.



Sketch space for the cross product idea: two non-parallel vectors span a plane, and $\mathbf{u} \times \mathbf{v}$ points perpendicular to that plane.